

Auteur: Anna Voronovska
Studierichting: Informatica
Oefeningenreeks 2D nr. 11.2

$$\begin{aligned}\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - 3}{x-3} &= \lim_{x \rightarrow 3} \frac{(\sqrt{2x+3} - 3)(\sqrt{2x+3} + 3)}{(x-3)(\sqrt{2x+3} + 3)} \\&= \lim_{x \rightarrow 3} \frac{(\sqrt{2x+3})^2 - 3^2}{(x-3)(\sqrt{2x+3} + 3)} = \lim_{x \rightarrow 3} \frac{2x+3-9}{(x-3)(\sqrt{2x+3} + 3)} = \lim_{x \rightarrow 3} \frac{2(x-3)}{(x-3)(\sqrt{2x+3} + 3)} \\&= \lim_{x \rightarrow 3} \frac{2}{\sqrt{2x+3} + 3} = \frac{2}{\sqrt{2 \cdot 3 + 3} + 3} = \frac{2}{6} = \frac{1}{3}\end{aligned}$$

References